Week 4 Discussion

Problem 1 (High-water marks: from a cdf to a pdf, and a change of units; C&B Exercise 1.53)

A certain river floods every year. Suppose that the low-water mark is set at 1 and the high-water mark Y has distribution function

$$F_Y(y) = P(Y \le y) = 1 - \frac{1}{y^2}, \qquad 1 \le y < \infty,$$

and $F_Y(y) = 0$ for $y < 1$.

(a) Verify that $F_Y(y)$ is a cdf, by checking the three conditions of Theorem 1.5.3. Is Y continuous, discrete, or neither?

(b) Find $f_Y(y)$, the pdf of Y. Find $P(Y > 2)$ twice: once from F_Y, and once by integrating f_Y.

(c) If the low-water mark is reset at 0 and we use a unit of measurement that is $\frac{1}{10}$ of that given previously, the high-water mark becomes $Z = 10(Y - 1)$. Find $F_Z(z)$. *Hint:* write the event $\{Z \le z\}$ as an event about Y.

(d) Find $f_Z(z)$ in two ways: by differentiating your answer to (c), and by Theorem 2.1.5. What are $\mathcal{Y}$ and $\mathcal{Z}$?

> **(a)** We check the three conditions of Theorem 1.5.3.
>
> - **Limits.** $F_Y(y) = 0$ for all $y < 1$, so $\lim_{y \to -\infty} F_Y(y) = 0$. Also $\lim_{y \to \infty} \left(1 - y^{-2}\right) = 1$.
> - **Nondecreasing.** On $[1, \infty)$, $\frac{d}{dy}\left(1 - y^{-2}\right) = 2y^{-3} > 0$, and F_Y is constant (0) for $y < 1$. At the junction, $F_Y(1) = 1 - 1 = 0$, so F_Y does not jump down.
> - **Right-continuous.** Each piece is continuous, and the two pieces meet at $y = 1$, since $\lim_{y \uparrow 1} F_Y(y) = 0 = F_Y(1)$.
>
> F_Y is in fact continuous everywhere, so Y is a **continuous** random variable (Definition 1.5.7).
>
> **(b)** $f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{2}{y^3}$ for $y > 1$, and $f_Y(y) = 0$ for $y < 1$.
>
> - From the cdf: $P(Y > 2) = 1 - F_Y(2) = \frac{1}{4}$.
> - From the pdf: $\int_2^\infty 2y^{-3}\, dy = -y^{-2}\Big|_2^\infty = \frac{1}{4}$.
>
> **(c)** Since $Z = 10(Y - 1)$, for $z \ge 0$,

$$F_Z(z) = P(Z \le z) = P\big(10(Y-1) \le z\big) = P\left(Y \le 1 + \frac{z}{10}\right)$$
$$= 1 - \frac{1}{(1+z/10)^2} = 1 - \frac{100}{(10+z)^2},$$

and $F_Z(z) = 0$ for $z < 0$.

(d)

- **Differentiating:** $f_Z(z) = \frac{d}{dz}\left(1 - 100(10+z)^{-2}\right) = \frac{200}{(10+z)^3}$, for $z > 0$.
- **Theorem 2.1.5:** $g(y) = 10(y-1)$ is increasing, with $g^{-1}(z) = 1 + z/10$ and $\frac{d}{dz} g^{-1}(z) = \frac{1}{10}$. So

$$f_Z(z) = f_Y\left(1 + \frac{z}{10}\right) \cdot \frac{1}{10} = \frac{2}{(1+z/10)^3} \cdot \frac{1}{10}$$
$$= \frac{2 \cdot 1000}{(10+z)^3} \cdot \frac{1}{10} = \frac{200}{(10+z)^3}.$$

- The supports are $\mathcal{Y} = (1, \infty)$ and $\mathcal{Z} = g(\mathcal{Y}) = (0, \infty)$.

Notice the factor $\frac{1}{10}$. Measuring in smaller units stretches the axis by a factor of 10, so the density must be 10 times lower for the total area to stay 1.

Problem 2 (A discrete transformation; C&B Exercise 2.3)

Suppose X has the geometric pmf $f_X(x) = \frac{1}{3}\left(\frac{2}{3}\right)^x$, $x = 0, 1, 2, \ldots$.

(a) Verify that f_X is a pmf (Theorem 1.6.5). Then find the cdf $F_X(x)$ for **every** real x. *Hint:* use the partial sum of the geometric series, as in (1.5.4).

(b) Determine the probability distribution of $Y = X/(X+1)$. Note that here both X and Y are discrete random variables. To specify the probability distribution of Y, specify its pmf.

- What is $\mathcal{Y}$?
- Is $g(x) = x/(x+1)$ one-to-one on $\mathcal{X}$? What is $g^{-1}(y)$?

(c) Find $P(Y \le 0.7)$.

(d) All the values of Y lie in $[0, 1)$, and they pile up near 1. Is Y still a discrete random variable? Explain using Definition 1.5.7.

(a) $f_X(x) \ge 0$, and

$$\sum_{x=0}^{\infty} \frac{1}{3}\left(\frac{2}{3}\right)^x = \frac{1}{3} \cdot \frac{1}{1-2/3} = 1.$$

For an integer $k \geq 0$, the partial sum gives

$$P(X \leq k) = \sum_{x=0}^{k} \frac{1}{3}\left(\frac{2}{3}\right)^x = \frac{1}{3} \cdot \frac{1-(2/3)^{k+1}}{1-2/3} = 1 - \left(\frac{2}{3}\right)^{k+1}.$$

Between integers the cdf is flat, so for every real x,

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1-(2/3)^{\lfloor x \rfloor + 1} & x \geq 0, \end{cases}$$

where $\lfloor x \rfloor$ is the largest integer $\leq x$. This is a step function, with a jump of size $f_X(k)$ at each $k = 0, 1, 2, \ldots$.

(b)

- $\mathcal{Y} = \{g(x) : x \in \mathcal{X}\} = \left\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots\right\} = \left\{\frac{k}{k+1} : k = 0, 1, 2, \ldots\right\}$.
- $g(x) = \frac{x}{x+1} = 1 - \frac{1}{x+1}$ is increasing, hence one-to-one. Solving $y = \frac{x}{x+1}$ gives $g^{-1}(y) = \frac{y}{1-y}$, so each $g^{-1}(y)$ is a single point.

The sum over $g^{-1}(y)$ has only one term, so

$$f_Y(y) = f_X\left(\frac{y}{1-y}\right) = \frac{1}{3}\left(\frac{2}{3}\right)^{y/(1-y)}, \qquad y = 0, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{3}{4}, \ldots,$$

and $f_Y(y) = 0$ otherwise. Equivalently, $P\left(Y = \frac{k}{k+1}\right) = \frac{1}{3}\left(\frac{2}{3}\right)^k$.

There is no Jacobian factor here. A discrete transformation only *relabels* the points: the probabilities move with them unchanged.

(c) Transform the event:

$$\{Y \leq 0.7\} = \left\{X \leq \frac{0.7}{0.3}\right\} = \{X \leq 2.33\ldots\} = \{X \leq 2\}.$$

So $P(Y \leq 0.7) = F_X(2) = 1 - (2/3)^3 = \frac{19}{27} \approx 0.704$. This is also Theorem 2.1.3(a): $F_Y(y) = F_X(g^{-1}(y))$ for increasing g, which does not require X to be continuous.

(d) Yes. $\mathcal{Y}$ is countable, and F_Y is a step function: it jumps by $\frac{1}{3}(2/3)^k$ at each $\frac{k}{k+1}$ and is flat in between. The jumps get closer together near 1, but there are still only countably many of them. Being "discrete" is about the cdf being a step function, not about the values being spread apart or being integers.

Problem 3 (When the pieces don't match: a piecewise-monotone transformation; C&B Exercise 2.7)

Let X have pdf $f_X(x) = \frac{2}{9}(x+1)$, $-1 \le x \le 2$, and let $Y = X^2$.

(a) Sketch f_X and the graph of $g(x) = x^2$ on $[-1, 2]$. What is $\mathcal{Y}$? Split $\mathcal{X}$ into the two pieces $(-1, 0)$ and $(0, 2)$ on which g is monotone, and find the image of each piece. Which condition of Theorem 2.1.8 fails? (This is why C&B note that Theorem 2.1.8 is not directly applicable in this problem.)

(b) Use the cdf method to find $F_Y(y)$. *Hint:* the event $\{X^2 \le y\} = \{-\sqrt{y} \le X \le \sqrt{y}\}$ looks different depending on whether $-\sqrt{y}$ is inside the support of X. Treat $0 \le y < 1$ and $1 \le y \le 4$ separately.

(c) Differentiate to find the pdf of Y. Check that it integrates to 1. Is it bounded?

(d) Show that Theorem 2.1.8 remains valid if the sets $A_0, A_1, \ldots, A_k$ *contain* $\mathcal{X}$, and apply the extension to solve part (c) using $A_0 = \emptyset$, $A_1 = (-2, 0)$, and $A_2 = (0, 2)$. *Hint:* outside the support, $f_X = 0$.

(a) $\mathcal{Y} = [0, 4]$.

- On $(-1, 0)$, g is decreasing, with image $(0, 1)$.
- On $(0, 2)$, g is increasing, with image $(0, 4)$.

The images differ, so condition **iii** of Theorem 2.1.8 (the same set $\mathcal{Y}$ for each piece) fails. Values $y \in (0, 1)$ have two pre-images, $\pm\sqrt{y}$. Values $y \in (1, 4)$ have only one, $+\sqrt{y}$, because $-\sqrt{y} < -1$ lies outside the support.

(b)

- $0 \le y < 1$**:** both endpoints $\pm\sqrt{y}$ lie in $[-1, 1]$, so

$$F_Y(y) = P(-\sqrt{y} \le X \le \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{2}{9}(x+1)\,dx$$
$$= \frac{2}{9} \cdot 2\sqrt{y} = \frac{4\sqrt{y}}{9}.$$

  The x term integrates to 0 over the symmetric interval.

- $1 \le y \le 4$**:** now $-\sqrt{y} \le -1$, so the left endpoint is cut off by the support:

$$F_Y(y) = P(-1 \le X \le \sqrt{y}) = \frac{1}{9}(x+1)^2\Big|_{-1}^{\sqrt{y}} = \frac{(\sqrt{y}+1)^2}{9}.$$

- $F_Y(y) = 0$ for $y < 0$ and $F_Y(y) = 1$ for $y > 4$.

Checks: both formulas give $\frac{4}{9}$ at $y = 1$, which is $P(-1 \le X \le 1)$. The second gives 1 at $y = 4$.

(c) Differentiating,

$$f_Y(y) = \begin{cases} \dfrac{2}{9\sqrt{y}} & 0 < y < 1 \\[2ex] \dfrac{1}{9}\left(1 + \dfrac{1}{\sqrt{y}}\right) & 1 < y < 4 \\[2ex] 0 & \text{otherwise.} \end{cases}$$

Check:

$$\int_0^1 \frac{2}{9\sqrt{y}}\,dy + \int_1^4 \frac{1}{9}\left(1 + y^{-1/2}\right)dy = \frac{4}{9} + \frac{1}{9}(3+2) = 1.$$

f_Y is **unbounded** near $y = 0$, yet it still integrates to 1. Compare Check your understanding 1 in Lecture 4.1.

(d) The proof of Theorem 2.1.8 only uses the pieces to split the event $\{Y \le y\}$ into parts on which g is monotone. Any x outside $\mathcal{X}$ has $f_X(x) = 0$, so it adds nothing to any integral. Enlarging the pieces to $A_1 = (-2, 0)$ and $A_2 = (0, 2)$ therefore changes nothing, and now each piece maps onto the same set $(0, 4)$, so condition iii holds. With $g_1^{-1}(y) = -\sqrt{y}$, $g_2^{-1}(y) = \sqrt{y}$, and $\left|\frac{d}{dy} g_i^{-1}(y)\right| = \frac{1}{2\sqrt{y}}$,

$$f_Y(y) = \frac{1}{2\sqrt{y}}\Big(f_X(-\sqrt{y}) + f_X(\sqrt{y})\Big), \qquad 0 < y < 4.$$

- For $0 < y < 1$: $\frac{1}{2\sqrt{y}} \cdot \frac{2}{9}\big((1 - \sqrt{y}) + (1 + \sqrt{y})\big) = \frac{2}{9\sqrt{y}}$.
- For $1 < y < 4$: $f_X(-\sqrt{y}) = 0$, because $-\sqrt{y} < -1$. So $f_Y(y) = \frac{1}{2\sqrt{y}} \cdot \frac{2}{9}(\sqrt{y} + 1) = \frac{1}{9}\left(1 + \frac{1}{\sqrt{y}}\right)$.

This matches (c). **Lesson:** the formula is the same everywhere. What changes with y is *which pre-images land inside the support*, which is exactly what the cdf method in (b) makes visible.